Answers retake exam
Exam in Introduction to economics for sport management

Instructions (read carefully)

a. The final exam lasts 3 hours.
b. You can use calculators.
c. Justify all your answers using economic concepts, theories and reasoning.

Good Luck!

Question 1 (20 points)

Suppose there is a country that can produce two goods: X and Y. The country has 400 machines and 200 workers. Each machine can produce 1 units of X or 2 units of Y. Each worker can produce 4 units of X or 4 units of Y.

a. Please draw a production possibilities frontier of this country and mark all the points (X and Y) where the curve changes its slope as well as the maximal production of X and Y. **(10 points)**
b. Suppose, there is a world trade, such that 1 unit of X is worth 3 units of Y. How many units of X and Y should this country produce and why? **(4 points)**
c. Suppose that this country needs to consume 1000 units of X. In that case how many units of Y will it consume? **(6 points)**

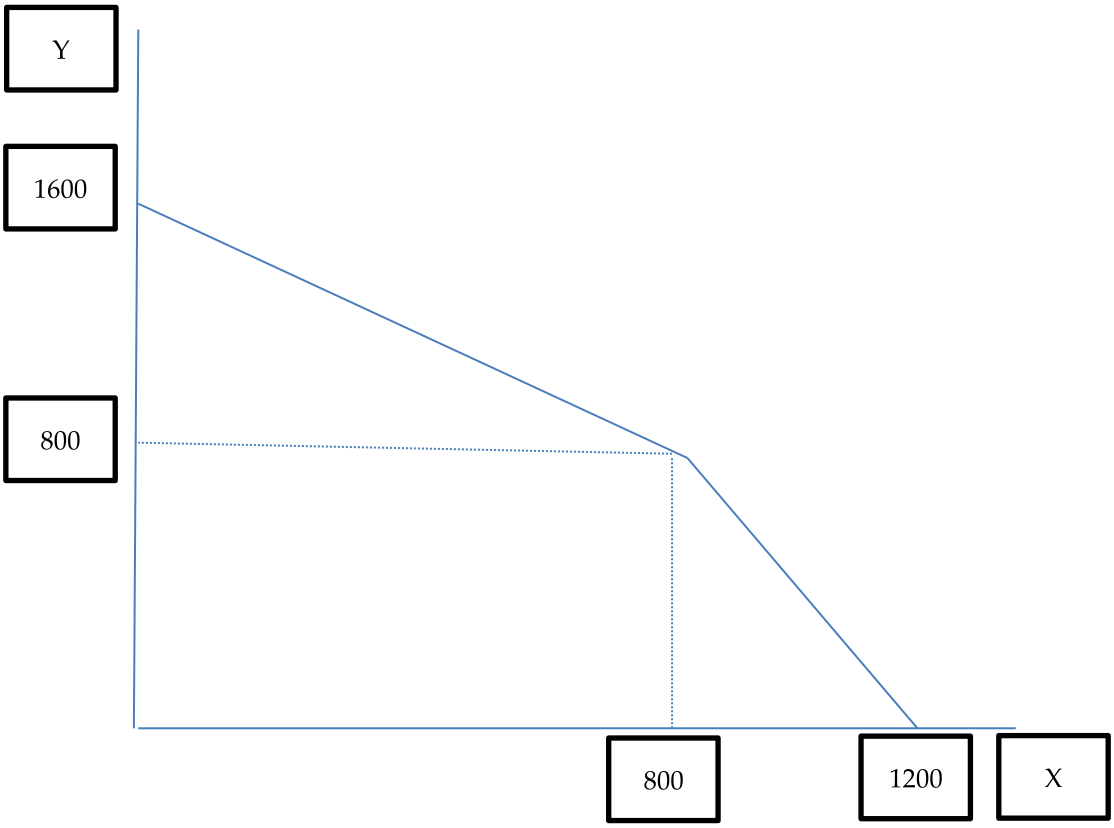
Answers:

	machines	workers
Production X per one unit	1	4
Production Y per one unit	2	4
MC in producing X in terms of Y	2	**1**
MC in producing Y in terms of X	**0.5**	1
Max X	400*1=400	200*4=800
Max Y	400*2=800	200*4=800

a. See answer a below.

b. This country has a comparative advantage in producing X, because in the worst case, its cost of producing X is 2Y. However they can sell it to the world for the price of 1X=3Y. Therefore, this country will only produce 1200X and zero Y.

c. If a country needs to consume 1000 units of X, then it will still produce 1200 units of X. The country will exchange the remaining 200 units of X for 200*3=600 units of Y.



Question 2 (20 points)
There is a market with 5 workers who should be allocated between two fields with the following production function:

Field A

L	TP
0	0
1	7
2	13
3	16
4	18
5	19

Field B

L	TP
0	0
1	4
2	7
3	9
4	10
5	10

a. How many workers would work on each field if it is known that the price of one produced unit on field A is 5USD and the price of one produced unit on field B is 1USD? **(6 points)**
b. What would be the salary of each worker? **(6 points)**
c. What would happen to the profit of the owner of field A if there will be 6 workers? **(8 points)**

Answer:

Field A

L	TP	mp	Vmp (p*mp)
0	0		
1	7	7	35
2	13	6	30
3	16	3	15
4	18	2	10
5	19	1	5

Field B

L	TP	mp	Vmp (p*mp)
0	0		
1	4	4	4
2	7	3	3
3	9	2	2
4	10	1	1
5	10	0	0

a. There will be 5 workers in Field A and zero workers in field B.
b. The value of the marginal product of the last worker is 5 and this will be the wage as well (VMP=W).

Additional worker will reduce the wage to 4, but the productivity of field A will remain the same. That means that the profit of the owner of field A will be increased.

Question 3 (20 points)

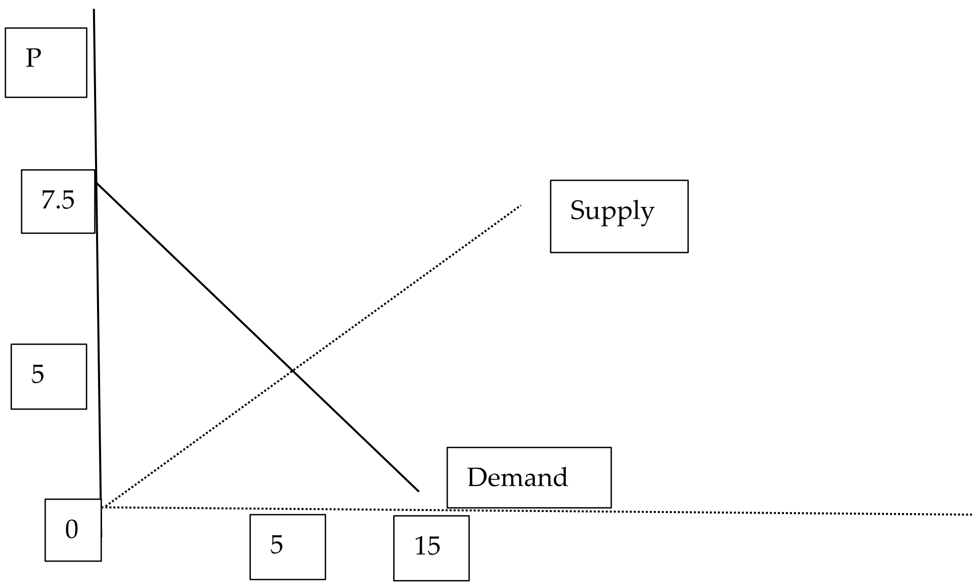
Suppose the following demand function: Q^D=15-2p. In addition suppose the following supply function: Q^S=p where Q^D is the quantity demanded of a good, Q^S is the supplied quantity of a good and p is the price of a good in USD.

a. What would be the price and the quantity in the equilibrium? **(4 points)**
b. Show on the graph (where appropriate) the cases where p=0, Q=0, and price and quantity in equilibrium. **(8 points)**
c. What is the producer surplus? **(3 points)**
d. What is consumer surplus? **(3 points)**
e. What is the total social welfare? **(2 points)**

Answer:

a. **In equilibrium the supply is equal to demand ($Q^S = Q^D$) therefore, 15-2p=p. Solving this equation we get p=5 and Q=5.**
b. **See the answer below**

c,d,e The total social welfare is the sum of consumer and producer surplus.
The consumer surplus is the area between the demand curve and the price (triangle area) (7.5-5)*(5-0)/2=6.25
The producer surplus is the area between the supply curve and the price (triangle area) (5-0)*(5-0)/2=12.5
The total social welfare is 6.25+12.5=18.75



Question 4 (16 points)

Assume a firm with the following production function:

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0				20			
1	8							
2	15							
3	27							
4	40							

It is known that one unit of labor (L) costs 20 USD.

a. Complete the table. **(8 points)**
b. What is the minimal price that the firm will produce in the short term? **(2 points)**
c. What is the minimal price that the firm will produce in the long term? **(2 points)**
d. It is known that the price is 240 USD. What would be the production of the firm? **(2 points)**
e. What would be its profit in the short term? **(1 points)**
f. What would be its profit in the long term? **(1 points)**

Answer:

TP (Q)	**L**	**MP**	**FC**	**VC**	**TC**	**MC**	**ATC**	**AVC**
0	**0**		**20**		**20**			
1	**8**	**1/8**	20	**160**	**180**	**160**	**180**	**160**
2	**15**	**1/7**	20	**300**	**320**	**140**	**160**	**150**
3	**27**	**1/12**	20	**540**	**560**	**240**	**186.7**	**180**
4	**40**	**1/13**	20	**800**	**820**	**260**	**205**	**200**

b. min AVC=150. Below that price the firm will not produce in the short term
c. min ATC=160. Below that price the firm will not produce in the long term
d. The rule of production is mc=p. Therefore, if p=240, the firm will produce 3 units of Q.
e. Profit (short)=P*Q-VC= 240*3-540=180
f. Profit (long)=P*Q-TC= 240*3-560=160

Question 5 (7 points)

You are expected to invest a one-time sum of 1000 NOK at the end of the second year from now and receive it at the end of the fifth year from now. What is the Future Value of this sum if the annual interest rate is 8% in the first year, 5% in the second year, 7% in the third year and 4% in the fourth and the fifth years?

Answer:

FV=1000*(1.04)^2*(1.07) =1157

Question 6 (7 points)

You were offered to invest in the business that will produce a cash flow of 1000 NOK at the end of each of the next three years. What is the Present Value of this cash flow if the annual interest rate is 1%?

Answer:

$PV=1000/(1.01)^3+1000/(1.01)^2+1000/(1.01)=2941$

Question 7 (5 points)

Suppose an elastic good X (elasticity above 1). Now suppose that the price of that good has decreased. What will happen to the total revenues of the suppliers?

Answer:

A decrease in the price of a good with elastic demand will increase the total expenditure of consumers (total revenue of suppliers).

Question 8 (5 points)

Provide a list of five possible gains from hosting Mega-Events.

Answer from the presentation (6 gains):

- **General infrastructure that are politically difficult to provide without the strict deadline of an opening ceremony (roads, airports, etc.).**
- **An increased number of tourists after the event.**
- **Promotion of a higher sports participation (health).**
- **New stadiums.**
- **Signal to investors that a country is ready to welcome foreign investments.**
- **"Putting a country on the map" (politically and economically).**